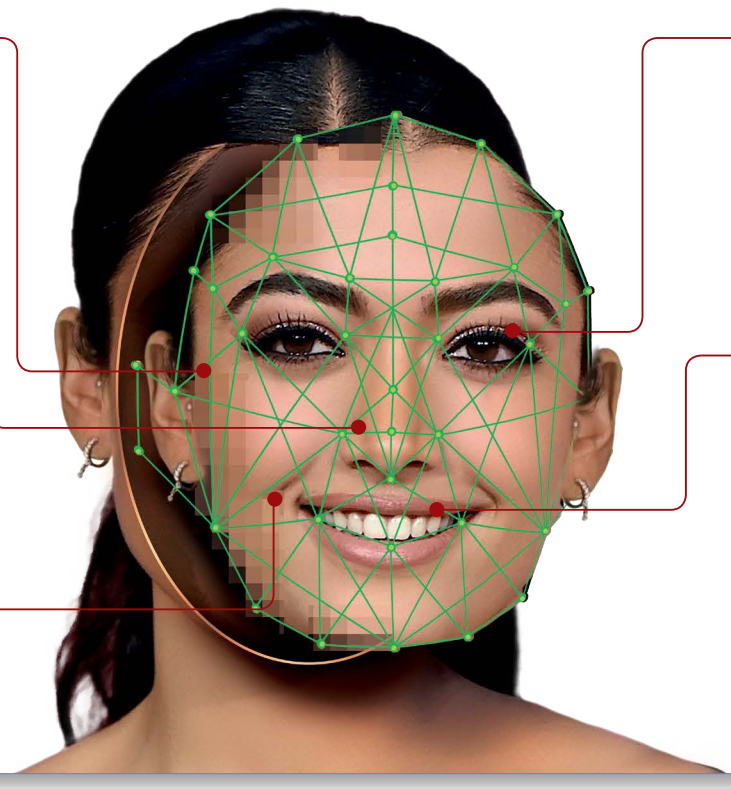
HOW TO DETECT A DEEPFAKE

With rapidly evolving tools, it's getting harder to detect deepfakes. However, there are some imperfections that act as obvious giveaways



SKIN
Look for cheeks and forehead with unnatural smoothness or wrinkles

SHADOWS AND REFLECTIONS
Unnatural lighting is another clue. Do the shadows cast by facial features appear natural? Does the angle of glare from glasses change with movements?

FACIAL EXPRESSIONS
Do they appear natural? Are they detailed or blurred?

BLINKING OF THE EYES
Eyes that don't blink were one of the early clues. But deepfake generators have got wiser since then. Look for unnatural blinking

LIP MOVEMENTS
If the video is fake, there could be unnatural lip movements and mismatch in voice syncing

Sources: US Homeland Security, MIT DetectFakes project, interviews

Illustrations by **NILANJAN DAS**

FACE SWAP ITSELF ISN'T NEW, WE HAD PHOTOSHOP IN THE '90S. DEEP LEARNING TECHNIQUES CHANGED THE GAME DRAMATICALLY

drasekhar, the Union minister of state for electronics and information technology (MeITY), says in the accompanying interview, "The phenomenon of deepfakes, which is really a marriage of AI and the misinformation industry, is certainly something that all of us should be worried about because it really is very potent for individuals, for societies, for communities and countries." No longer is it merely a matter of discerning the real from the synthetic. In the wrong hands, it's potentially a dangerous subversion of anything that's real.

HOW DEEP IS THE PROBLEM

Since their emergence, the developments in deepfake technologies have been rapid, and the costs of producing top-quality deepfake content declining. Besides celebrities, top global figures—from Pope Francis to former American presidents Barack Obama and Donald Trump—have been the subjects of relentless memes. On November 17, Prime Minister Narendra Modi, warning that deepfakes could lead to a crisis and 'stoke the fire of discontent' in society, revealed how he recently saw a deepfake of himself doing the Gujarati 'garba', a dance he hadn't performed since school.

A paper by global consulting firm KPMG estimates that there are upwards of 100,000 AI-based models to create or detect deepfakes. Of these, only about 3,000 are 'able to identify deepfakes properly', most of them being able to discern only the cruder fakes. According to Sensity.ai, a Netherlands-headquartered visual threat intelligence company, the number of fake videos online has been growing exponentially since 2018, roughly doubling every six months. The firm says it detected 85,047 fake videos online as of December 2020.

And, it's not confined just to video. In 2019, the story of a deepfake audio used to steal $243,000 from the UK subsid-